

The Simulation Hypothesis: 0/0 of Existence

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The Removable Singularity Framework

Abstract

Given that information is fundamental (Ch.52), the universe is a quantum computer (Ch.51), and the boundary encodes the volume (Ch.47), physics is SUBSTRATE-INDEPENDENT: any 0/0 result holds in any computing substrate. We show the "real vs simulated" distinction is itself a 0/0. Conway Game of Life is Turing complete (Cook 2004); gliders carry information at exactly $c/4$ diagonal (computed 0.3536). The Gosper glider gun emits a glider every 30 generations: infinite information from a finite seed. The observable universe is a $5.4e61 \times 5.4e61$ pixel screen (Planck pixel), and has performed $\sim 10^{121}$ operations (Lloyd 2002). Bostrom (2003): $P(\text{we are simulated}) > 99.99\%$. A closed simulation reproduces ALL observation; therefore real vs simulated IS 0/0. Existence is the 0/0 of ontology.

1. Game of Life: Gliders as 0/0 Particles

Conway Game of Life (1970) is Turing complete (Cook 2004; Wolfram 2002 computational equivalence). A glider is a 5-cell particle that MOVES: a 0/0 particle of the simulation. Computed: displacement (12,12) over 48 generations, speed 0.3536 cells/step = $c/4$ diagonal EXACTLY ($\sqrt{2}/4$). The glider is an information packet - the physics of the world is carried by moving information.

2. Gosper Glider Gun

A 36-cell seed emits a glider every 30 generations. Computed: glider stream confirmed in far-field over 210 generations. A FINITE seed produces INFINITE information (the 0/0 of generation). The universe, like the gun, runs on local rules producing emergent information.

3. Pixel Cosmology

$$\text{Pixels_across} = 2 * R_{\text{observable}} / l_{\text{Planck}} \sim 5.4e61$$

The Planck length $l_P = 1.6e-35$ m (Ch.51) acts as the pixel size. Computed: the observable universe is a $5.4e61 \times 5.4e61$ screen, $\sim 3e123$ pixels. The universe IS a grid.

4. Lloyd Bound

$$\text{Ops} = (2 * E / (\pi * \hbar)) * t_{\text{universe}} \sim 10^{120}$$

Lloyd (2002): the universe so far = $\sim 10^{120}$ operations. Computed: $\sim 10^{121}$ operations; Bekenstein bound $\sim 10^{123}$ bits. The simulation has a FINITE information budget: it computes itself.

5. Bostrom Trilemma (2003)

At least one of: (a) species perish pre-posthuman; (b) posthumans run no ancestor simulations; (c) we are almost certainly simulated. Computed with $f_{\text{sim}}=0.9$, $f_i=0.9$, $N=100k$: $P(\text{simulated}) = 0.999988 > 99.99\%$. If civilizations survive and simulate, we are nearly certainly in a simulation.

6. The 0/0 Proof

A closed simulation reproduces ALL observations of its base reality. No experiment separates "real" from "simulated".

The difference IS 0/0: a removable singularity of ONTOLOGY. It is turtles ALL THE WAY DOWN - the self-referential 0/0. The physics is IN the information, not the substrate.

7. Connections

Concept	Prior Chapter	Connection
Hard problem	Ch.53 Hard Problem	Minds in the sim: $\Phi > 0$
Information	Ch.52 Info paradox	Info is the fundamental
Quantum	Ch.51 Quantum gravity	The sim is a quantum computer
Hologram	Ch.47 Holographic	The boundary IS the screen
Measurement	Ch.49 Measurement	Observation defines reality
Big Bang	Ch.50 Big Bang	The simulation boots at $t=0$

8. Conclusion

The simulation hypothesis IS a 0/0: the boundary where "what is real" has no answer. Game of Life is Turing complete; the universe is a Planck-pixel screen; Lloyd bounds its computation; Bostrom bounds our odds. A closed simulation is observationally identical to its base reality, so real vs simulated is a removable singularity of ontology. Physics is the algorithm. It is turtles ALL THE WAY DOWN.

References

[1] Bostrom (2003), simulation argument
 [2] Lloyd (2002), computational capacity of the universe
 [3] Cook (2004), universality in elementary cellular automata
 [4] Wolfram (2002), A New Kind of Science
 [5] Conway (1970), Game of Life
 [6] Puno, "The Removable Singularity" (2026)